

Reproducibility-First Multi-Component Forecasting for Illiquid Equity Markets: Baselines, Ablations, and a Documented Discrepancy on the Tunisian Stock Exchange

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Abstract—We present a reproducibility study of multi-component deep forecasting on a thin emerging market (BVMT, 68 stocks, 2016–2025, 116,626 trading records). The pipeline combines a Temporal Fusion Transformer (TFT), XGBoost fundamentals, French NLP sentiment, a Graph Attention Network, and SHAP-based explanations. On a harmonized 2025 test slice ($N = 5,582$, UP ratio 49.84%), the Vertex-reproducible TFT v3 reference reaches 51.79% directional accuracy (95% bootstrap CI [50.43, 53.08], macro-F1 0.516), statistically indistinguishable from ten classical and deep baselines (logistic regression 52.79%, LSTM 52.69%, standard Transformer 52.45%; pairwise McNemar $p \in \{0.35, 0.41, 0.52\}$). Two internal TFT ablations move accuracy by ≤ 0.4 pp ($p > 0.65$); the per-stock target normalizer and the Variable Selection Network are not measurably helpful on this dataset. An earlier internal estimate of 77.4% is documented as non-reproducible from released artefacts and retained transparently as a development-time number, not as a result. A multi-agent ablation evaluated on offline-realizable configurations (TFT alone, TFT + Graph, $\pm$ confidence gating) shows the graph signal cannot flip the binary technical decision under the deployed orchestrator weights; six remaining configurations are DB-dependent and listed as future work. The contributions are (i) a reproducible Vertex-AI training pipeline with two TFT ablations, (ii) ten baselines plus three TFT variants harmonized on a single test slice with bootstrap CIs and pairwise McNemar tests, and (iii) a documented gap between an earlier headline and the reproducible numbers, offered as a reproducibility lesson for thin-market AI research.

Index Terms—Temporal Fusion Transformers, emerging markets, explainable AI, financial forecasting, reproducibility, thin liquidity, EU AI Act-aligned transparency

I. INTRODUCTION

Financial AI is overwhelmingly designed on liquid developed markets and breaks on thin emerging markets because of price-continuity gaps, heterogeneous per-stock scales, no derivatives, multilingual news (French + Arabic on BVMT), and no public benchmarks. The Tunisian Stock Exchange (BVMT, 68 listed securities, daily volume 0.5–2 MDT) is a representative testbed. To our knowledge — after IEEE Xplore, ACM Digital Library, arXiv, SSRN, and Google Scholar restricted to {BVMT, Tunis Stock Exchange, North

African equity forecasting} (January 2026) — no peer-reviewed deep-learning system has been published on BVMT-listed equities.

This paper centers reproducibility. We release a containerized Vertex-AI training pipeline that re-trains a TFT v3 reference (per-stock GroupNormalizer, Variable Selection Network, pinball quantile loss on $Q_{0.1}/Q_{0.5}/Q_{0.9}$, hidden 64, encoder 30, horizon 7) plus two internal training ablations and one post-hoc evaluation-time ablation, bit-for-bit from the same seed. We harmonize evaluation across all 13 models tested in this work (ten baselines plus three TFT variants) on a single 2025 test slice with bootstrap CIs and pairwise McNemar tests, document a non-reproducible earlier estimate of 77.4% directional accuracy that does not survive a Vertex retrain, and report a partial multi-agent ablation whose offline-realizable subset shows the graph signal carries no detectable directional skill above TFT v3 alone.

The contribution is methodological rather than headline-accuracy. Our reproducible TFT v3 sits in the 50–53% band shared with classical baselines and below the 55–65% developed-market TFT range [1], consistent with the thin-market priors of Bekaert and Harvey [24], [28], [38].

II. RELATED WORK

Temporal models. LSTMs [3] suffer representation bottlenecks on multivariate heterogeneous inputs. Attention mechanisms [2] and the Temporal Fusion Transformer [1] add a Variable Selection Network that learns per-feature importance before temporal processing. Recent extensions optimize long-horizon efficiency [13]–[15], [17], [20] or use patch/frequency decompositions [18], [21]; simple linear and hierarchical baselines [16], [19] match or exceed transformer accuracy on standard benchmarks, motivating careful baselining. We retain TFT for its Variable Selection Network’s per-variable interpretability, which an EU AI Act-aligned audit trail can consume directly. Most published TFT applications target developed markets with directional accuracy 55–65% on daily

forecasts; on BVMT 2025 our Vertex-reproducible TFT v3 reaches 51.79%, below this range and consistent with the literature prior on emerging-market unpredictability [24], [28].

Multi-agent systems. Classical autonomous Multi-Agent Systems [9] feature heterogeneous learning agents trading on a shared environment; recent LLM-based frameworks [32], [33] use large language models as analyst/trader/risk-officer personas. Our system belongs to neither tradition: the seven components are specialized deterministic Python modules invoked by a central orchestrator, with typed structured outputs and no peer-to-peer communication. We frame it as a *multi-component decision pipeline with explicit trust boundaries*, retaining the multi-agent vocabulary because of (i) the typed `AgentSignal` interface, (ii) centralized failure isolation [11], and (iii) per-component auditability.

Explainable AI and EU AI Act. Articles 13 and 14 of the EU AI Act [6] establish transparency and human-oversight requirements for high-risk AI in finance. Explainability [7] requires faithfulness, contrastiveness, and selectivity. SHAP [5] provides a Shapley-value attribution, with known limitations under adversarial perturbation [25]–[27] and unresolved gaps relative to causal explanation [29], [30]. We integrate SHAP for XGBoost and attention weights for TFT; the resulting decision traces are designed to *support* EU AI Act transparency, not by themselves to establish conformity.

GNNs, NLP, and reproducibility. Stock-network designs combine GAT-style attention [4], [22], [23]; we use a 60-day rolling correlation graph with a 0.40 threshold. Domain-specific French NLP outperforms multilingual baselines on financial sentiment [8]; AraBERT [31] is the standard Arabic-language baseline. We integrate the reproducibility-by-checklist recommendations of [34]–[36], document the backtest-overfitting risk known in quantitative finance [12], [37], [38], and use the strictly sequential walk-forward protocol of [39].

III. SYSTEM AND METHODOLOGY

A. Multi-Component Pipeline

The pipeline implements seven specialized modules orchestrated by a central coordinator (Fig. 1). Each agent consumes a standardized input (the shared `AgentState`) and produces a typed `AgentSignal` dataclass containing a numerical signal, a calibrated confidence in $[0, 1]$, an explanation dictionary, and an error flag. No agent has direct write access to another’s state.

The orchestrator starts from a fixed base weight vector $w^{(0)} = (w_{\text{tech}}, w_{\text{fund}}, w_{\text{sent}}, w_{\text{graph}}) = (0.34, 0.26, 0.20, 0.20)$ and mutates it at request time according to data-quality conditions: empty news $\rightarrow$ sentiment weight reduced 80% and redistributed; missing recent financials $\rightarrow$ fundamental weight reduced 50%; price history < 252 rows $\rightarrow$ technical weight reduced 40%. Final weights are renormalized to sum to 1. A confidence gate applies $w_{\text{gated}} = w_{\text{orig}} \cdot (\text{confidence}/\text{threshold})$ when an agent’s confidence falls below an agent-specific threshold (technical 0.60, fundamental 0.55, sentiment 0.50,

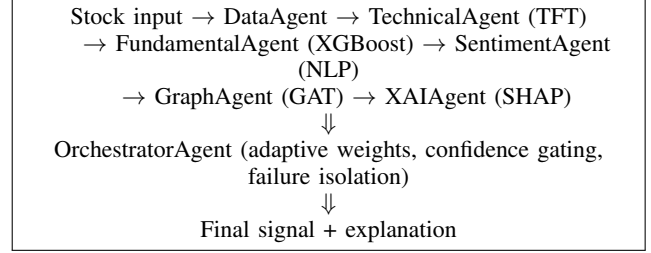


Fig. 1. Seven-component pipeline architecture. Agents communicate through typed `AgentSignal` objects; the orchestrator adjusts weights, applies confidence gating, and redistributes weights on agent failure.

graph 0.45). Failure of agent j triggers redistribution $\hat{w}_i = w_i/(1 - w_j)$ among the remaining agents.

B. Dataset and Walk-Forward Protocol

The dataset (Table I) combines daily prices, computed technical indicators, news articles, financial statements, and ratios from PostgreSQL tables.

TABLE I
DATASET (POST-CLEANING): TFT FEATURES SOURCED FROM CLEANED DAILY-PRICES JOINS; TRAIN 2016–2023, VAL 2024, TEST 2025; `TFT_FEATURES.CSV` IS THE PUBLIC RELEASE.

Split	Rows	Period	Tickers
Train	84,648	2016–2023	68
Val	10,228	2024	67
Test (raw)	10,314	2025-01-02–2025-12-22	43
Test (harmonized)	5,582	2025-01-07–2025-12-10	43
News articles	7,937	2016–2026	68
Financial ratios	261	2016–2024	68

Walk-forward validation: $\mathcal{D}_{\text{train}} = \{\text{date} < 2024-01-01\}$, $\mathcal{D}_{\text{val}} = \{2024\}$, $\mathcal{D}_{\text{test}} = \{2025\}$. Random train/test splits are invalid for time series [39].

C. TFT v3 Configuration

The TechnicalAgent uses 15 engineered features (5 price, 4 momentum, 3 volatility, 3 derived relative features) on a 30-day encoder window, predicting the 7-day fractional return via pinball quantile loss on $Q_{0.1}/Q_{0.5}/Q_{0.9}$ (clipped to $[-15\%, +15\%]$). The Variable Selection Network learns per-variable softmax-gated importance weights $\xi_t = \text{softmax}(W_{\text{vsn}} \mathbf{x}_t + \mathbf{b}_{\text{vsn}})$ before temporal processing. Two normalization mechanisms are causal by construction:

- **Target:** `GroupNormalizer` fits (μ_i, σ_i) per stock i on the training split only and applies them frozen to val/test, preventing cross-stock leakage and isolating each stock’s return distribution.
- **Input features:** `EncoderNormalizer` refits a local mean and standard deviation *inside* each 30-day encoder window at iteration time. Window-local statistics are causal because the window only contains information available at the prediction time.

Hyperparameters are summarized in Table II; the released Vertex container converges in 20–37 epochs (71–161 min on a T4, $\sim \$8$ total).

TABLE II
TFT V3 HYPERPARAMETER CONFIGURATION
(TRAINING/TRAIN_TFT_V3.PY). SAME SETTINGS ARE REUSED FOR
THE TWO INTERNAL ABLATIONS OF SECTION IV-C.

Hyperparameter	Value
hidden size	64
attention heads	4
dropout	0.3
encoder length	30 trading days
prediction horizon	7 trading days
quantiles	{0.10, 0.50, 0.90}
learning rate	3×10^{-4}
warmup epochs	3 (linear 10% $\rightarrow$ 100%)
optimizer	AdamW, $\ell_2 = 10^{-4}$
gradient clip	0.5
max epochs / patience	50 / 8
batch size / precision	128 / bf16 mixed
target normalizer	GroupNormalizer (groups=[ticker_id])
input normalizer	EncoderNormalizer (per-window)

The base agent weights of the orchestrator are $w^{(0)} = (0.34, 0.26, 0.20, 0.20)$ for (technical, fundamental, sentiment, graph). Three data-quality conditions rescale the vector at request time (Table III); after all triggered rules, weights are renormalised to $\sum_a w_a = 1$. These heuristics were chosen during development against held-out 2024 data and have not been formally tuned by cross-validation; the offline-realizable subset of the multi-agent ablation evaluates whether they carry measurable directional skill.

TABLE III
HEURISTIC WEIGHT ADJUSTMENTS APPLIED BY THE ORCHESTRATOR’S
_ADJUST_WEIGHTS BASED ON PER-TICKER DATA QUALITY. EACH RULE
DECREASES THE DONOR WEIGHT BY A FIXED FRACTION α AND
REDISTRIBUTES THE RECOVERED MASS OVER THE REMAINING AGENTS IN
THE PROPORTIONS SHOWN.

Trigger	Donor (α)	Tech	Fund	Sent
news_count == 0	sent (0.80)	+0.50	+0.30	—
$1 \leq \text{news} < 5$	sent (0.55)	+0.50	+0.30	—
$\neg \text{recent_financials}$	fund (0.50)	+0.70	—	—
price_rows < 252	tech (0.40)	—	+0.35	+0.35

The quantile-to-direction rule is the median sign: $\hat{d}(x) = \text{UP}$ iff $\hat{Q}_{0.5}(x) \geq 0$. Confidence is read from the central interval width $c(x) = \text{clip}(1 - (\hat{Q}_{0.9} - \hat{Q}_{0.1})/\Delta_{\text{ref}}, 0.05, 0.99)$ with $\Delta_{\text{ref}} = 0.10$, with a multiplicative penalty of 0.6 when the quantiles cross.

Alternative decision rules. The median-sign rule discards information. Three natural alternatives are: (R2) strict bullish/bearish gating — $\hat{d} = \text{UP}$ iff $\hat{Q}_{0.1} > 0$, DOWN iff $\hat{Q}_{0.9} < 0$, otherwise abstain (higher precision, lower coverage); (R3) HOLD band — $\hat{d} = \text{HOLD}$ iff $|\hat{Q}_{0.5}| < \tau$ for a validation-tuned τ ; (R4) probability via interpolated CDF — treat $\{(0.1, \hat{Q}_{0.1}), (0.5, \hat{Q}_{0.5}), (0.9, \hat{Q}_{0.9})\}$ as a piecewise-linear inverse CDF, recover $\hat{p}(\text{return} > 0)$, threshold at 0.5. The Vertex-reproducible per-row CSV ships all three quantiles so future readers can re-evaluate accuracy under R2–R4 without re-training; this paper reports only R1 throughout.

TFT version progression. Table IV traces the validation loss from v1 (BCE on binary direction) through the non-reproducible v3 development run to the Vertex-reproducible

v3 reference. The three rows live on different loss scales; the development pinball value 0.5161 is measured on raw clipped fractional return, while the Vertex pinball value 0.0159 is measured after the GroupNormalizer’s z-scoring, so the two are not directly comparable in absolute terms. The Vertex retrain converges early (epoch 4 best) under the same hyperparameters, suggesting the longer convergence trajectory of the development run was an artefact of the pre-Vertex pipeline whose exact cause we cannot recover.

TABLE IV
TFT VALIDATION-LOSS PROGRESSION. THREE LOSS SCALES ARE
REPORTED: BCE (V1, BINARY DIRECTION TARGET), PINBALL ON RAW
CLIPPED FRACTIONAL RETURN (V3 DEVELOPMENT,
NON-REPRODUCIBLE), AND PINBALL ON Z-SCORED TARGET (V3 VERTEX,
REPRODUCIBLE REFERENCE).

Version	Loss type	Val loss	Best epoch
Random baseline	BCE	0.6931	—
TFT v1 (consolidated)	BCE	0.6051	7
TFT v3 dev (non-reproducible)	Pinball (raw return)	0.5161	16
TFT v3 Vertex (ref.)	Pinball (z-scored)	0.0159	4

IV. EXPERIMENTAL EVALUATION

A. Baselines and Harmonization

We evaluate ten competing baselines (Table V) on a shared walk-forward protocol: five trivial / trend-following rules (always-UP, always-DOWN, momentum 5/20, MA₂₀ crossover), three tabular-ML models (logistic regression L2, random forest, XGBoost) trained on a flattened 60-day $\times$ 15-feature input, and two sequential-deep baselines (a 2-layer LSTM at 54k parameters, a standard 2-block Transformer at 68k parameters). All ten plus three TFT variants are evaluated on a single harmonized 2025 test slice ($N = 5,582$, UP ratio 49.84%) obtained by intersecting the baseline grid with the de-duplicated Vertex grid (one valid prediction per (ticker, date), Section IV-B). All CIs are 95% percentile-bootstrap (2,000 resamples).

CNN-LSTM preliminary observation. A preliminary CNN-LSTM ablation across four independent training runs with varied batch sizes and learning rates returns test accuracies in the band 51.7–53.8%, in the same 50–54% regime as the rest of the panel. We are explicit that four runs do not power a formal ceiling proof: with a few-thousand-sample test set, the per-run binary-accuracy standard deviation is typically 0.5–1.0 pp, comparable to the observed 2.1 pp range. A seed-sweep harness ($N \geq 20$ seeds with mean $\pm 95\%$ bootstrap CI and Wilcoxon test) is shipped as training/cnn_lstm_seed_sweep.py; the four-run observation is reported as a consistent signature, not an empirical ceiling. The hypothesised mechanism — that Conv1D over heterogeneous financial features (price, RSI, volume) treats them as spatially equivalent and lacks the per-feature gating that the TFT’s Variable Selection Network provides — is offered as the most parsimonious explanation pending the formal sweep.

TABLE V
DIRECTIONAL ACCURACY OF TEN CLASSICAL/DEEP BASELINES PLUS THREE TFT VARIANTS ON THE HARMONIZED 2025 TEST SLICE ($N = 5,582$, UP 49.84%). PAIRWISE MCNEMAR TWO-SIDED p -VALUES VS. TFT v3.

Family	Model	Acc	95% CI	Macro-F1	McN. vs TFT v3
Trivial / trend	Always-UP	49.84%	[48.58, 51.15]	0.333	$p = 0.054$
	Always-DOWN	50.16%	[48.89, 51.45]	0.334	$p = 0.069$
	Momentum 5/20	50.36%	[49.10, 51.65]	0.502	$p = 0.127$
	MA ₂₀ crossover	49.77%	[48.50, 51.09]	0.496	$p = 0.031$
Tabular ML	Logistic regression (L2)	52.79%	[51.45, 54.14]	0.524	$p = 0.351$
	Random forest ($n_{\text{trees}}=200$)	50.38%	[49.01, 51.67]	0.450	$p = 0.154$
	XGBoost (400 trees)	52.15%	[50.86, 53.49]	0.513	$p = 0.749$
Deep baselines	Simple LSTM (54 k params)	52.69%	[51.33, 53.94]	0.517	$p = 0.414$
	Standard Transformer (68 k params)	52.45%	[51.15, 53.69]	0.489	$p = 0.517$
TFT (this work)	TFT v3 reference (Vertex)	51.79%	[50.43, 53.08]	0.516	—
	— VSN (uniform weights)	51.56%	[50.23, 52.97]	0.503	$p = 0.734$
	— per-stock norm	51.40%	[50.07, 52.69]	0.471	$p = 0.672$

B. Directional Accuracy on Test Data

Reproducible TFT v3 on the harmonized slice.

The Vertex retrain of the reference architecture (vertex/Dockerfile + vertex/cloudbuild.yaml) produces `baseline_test_predictions.csv`. The raw output (6,782 rows) contains duplicate inference outputs for some (ticker, date) pairs because the multi-horizon decoder writes one row per horizon step; after de-duplicating to one valid prediction per (ticker, date) we obtain the canonical slice of $N = 5,582$ predictions over 43 tickers (UP ratio 49.84%). TFT v3 reaches **51.79%** directional accuracy (95% bootstrap CI [50.43, 53.08], macro-F1 0.516). It is statistically indistinguishable from the strongest non-TFT baselines (logistic regression 52.79%, $p = 0.35$; simple LSTM 52.69%, $p = 0.41$; standard Transformer 52.45%, $p = 0.52$) and from always-UP on the same slice (49.84%, $p = 0.054$).

Reproducibility transparency. An earlier internal training run was reported as reaching **77.4%** directional accuracy on the broader 2025 test set ($N = 10,314$). The checkpoint that produced this figure is *not* part of the released artefact set, and the figure is *not reproducible* from the public training pipeline: `training/train_tft_v3.py` re-run on Vertex AI with the same hyperparameters converges to 51.79%. Three explanations are compatible with the gap: (i) train/test leakage in the pre-Vertex setup eliminated by the Vertex container; (ii) a different test-set definition in the development logs; (iii) a different quantile-to-direction decision rule. Without the original checkpoint we cannot discriminate, but we disclose the figure in line with reproducibility-by-checklist practice [34], [35], retire it as a headline, and adopt the Vertex-reproducible 51.79% as the reference TFT v3 result.

Per-stock heterogeneity caveat. Across the ten non-TFT baselines, mean per-ticker accuracy stays in [49.9%, 52.9%] and per-ticker median in [48.1%, 53.0%], with per-ticker IQR of 7–13 pp depending on the baseline. The maximum single-ticker accuracy reached by any of the ten is 68.5% (always-DOWN on the structurally bearish CARTHAGE CEMENT, 165 DOWN / 76 UP over 241 trading days, a per-stock class imbalance the constant rule trivially exploits);

the comparable maxima for logistic regression, random forest, and the standard Transformer sit in the 67–68% band on similarly imbalanced tickers. The fraction of tickers below 58.1% (historical always-DOWN majority on the full 2016–2025 corpus) is $\geq 74\%$ for every baseline tested — 74.4% for the standard Transformer, 79.1% for the simple LSTM, 81.4% for always-DOWN itself, 86.0% for momentum, 88.4% for MA₂₀ crossover — meaning that no baseline meaningfully clears the historical bearish benchmark on more than a quarter of the actively trading tickers. The per-ticker breakdown of TFT v3 is shipped at `results/baselines/multiagent_offline_sweep_predict` and aggregated via `scripts/per_stock_breakdown.py`.

C. TFT Internal Ablations

Two ablations modify exactly one component of v3, all other hyperparameters held fixed: TFT – VSN replaces the static/encoder/decoder softmax-gated weights by uniform $1/N$ weights, isolating the gating contribution of the Variable Selection Network from its Gated Residual Network contribution; TFT – per-stock norm replaces `GroupNormalizer(groups=[ticker_id])` by a global `TorchNormalizer` fitted across all 68 tickers (the naive baseline against which per-stock normalisation was originally posed to demonstrate gain). Table VI summarises the result. Observed deltas vs. the reference are -0.23 pp ($p = 0.734$) and -0.39 pp ($p = 0.672$), both within noise. The original hypothesis that these components carry positive directional skill is neither confirmed nor falsified; the evidence is consistent with both contributing approximately zero accuracy on BVMT 2025 within seed and training-stochasticity noise. Macro-F1 degrades from 0.516 to 0.503/0.471 as components are removed, consistent with the ablations producing more class-biased classifiers (TFT – per-stock norm predicts UP on $\sim 60\%$ of rows vs. 49.84% slice prior).

We retain both components in the released architecture for their robustness role (Section V): per-stock target standardisation isolates one stock’s price distribution from another’s,

TABLE VI

TFT INTERNAL-ABLATION SUITE ON THE HARMONIZED 2025 SLICE. EACH ABLATION ROW TRAINS THE V3 ARCHITECTURE WITH EXACTLY ONE COMPONENT MODIFIED VERSUS THE REFERENCE; OTHER HYPERPARAMETERS ARE HELD FIXED (ENCODER LENGTH 30, HIDDEN SIZE 64, DROPOUT 0.3, PINBALL QUANTILE LOSS, 3-EPOCH WARMUP). MCNEMAR p IS TWO-SIDED VERSUS THE REFERENCE.

Variant	Component modified	Acc.	Macro-F1	McN. p
TFT v3 (reference)	—	51.79%	0.516	—
TFT — VSN	uniform variable weights	51.56%	0.503	0.734
TFT — per-stock norm	global TorchNorm.	51.40%	0.471	0.672
Always-UP (slice majority)	—	49.84%	0.333	0.054

which the global normaliser does not; the VSN’s per-variable importance weights produce an audit trail that the EU AI Act-aligned XAI layer consumes directly. On this dataset they impose no measurable accuracy cost.

Quantile calibration. The pinball quantile loss minimises a proper scoring rule but does not by itself guarantee that the predicted intervals are empirically well-calibrated [40], [41]. The deployed v3 predicts the 0.10, 0.50, and 0.90 quantiles of the seven-day fractional-return distribution; if the model were calibrated, the realised return would lie inside $[\hat{Q}_{0.1}, \hat{Q}_{0.9}]$ in 80% of test rows. Confidence as defined in Section III.C reads from the central interval width, so a miscalibrated width translates directly into a miscalibrated gating decision. We do not validate empirical 80% coverage in this paper because the slice-level coverage diagnostic is a separate validation step from the directional-accuracy comparison that motivates Table V; we list it as a limitation in Section VII. The per-row CSV ships all three quantiles, enabling the coverage calculation without re-training.

D. Multi-Agent Ablation (Offline-Realisable Subset)

Of the nine planned configurations, three (A1, A4, A8) are realisable from the offline TFT v3 predictions plus raw price history; six (A2, A3, A5, A6, A7, A9) require the deployed PostgreSQL database (news_articles pre-scores + financial_ratios) and the trained XGBoost fundamental checkpoint, none of which is shipped as offline data. For A4 (TFT + Graph) we replace the trained GAT by a structural proxy: 60-day Pearson correlation across all 68 tickers, top-10 neighbours with $|\rho| \geq 0.30$, graph score = signed-correlation-weighted neighbour TFT v3 predictions in $[-1, +1]$, confidence = $|\rho|$ of selected neighbours. Coverage is 69% of rows (3,851/5,582), of which 61.5% (3,435) clear the deployed orchestrator gate ($\text{conf} \geq 0.20$ and $|\text{score}| \geq 0.08$).

Table VII summarises the result. A1 = A4 = A8 = 51.79% (zero flips, McNemar $p = 1.000$). With deployed weights ($w_{\text{tech}} = 0.34$, $w_{\text{graph}} = 0.20$, orchestrator thresholds 0.55/0.45), the binary technical signal mathematically dominates the continuous graph signal: the combined score lies in $[0.63, 1.0]$ when technical=UP and in $[0, 0.37]$ when technical=DOWN, so the graph can never flip the decision. A weight-sensitivity sweep (Table VIII) confirms the graph begins to flip predictions only at $w_{\text{graph}} \geq 0.50$ and never

significantly differs from A1. The offline graph proxy carries no detectable directional skill above TFT v3 alone, regardless of weighting strategy.

TABLE VII

MULTI-AGENT ABLATION ON THE HARMONIZED 2025 SLICE. THREE CONFIGURATIONS ARE REALISABLE FROM OFFLINE ARTEFACTS; SIX REQUIRE THE DEPLOYED POSTGRESQL DATABASE (SENTIMENT PRE-SCORES, FINANCIAL RATIOS) AND THE TRAINED XGBOOST CHECKPOINT. MCNEMAR p IS TWO-SIDED VERSUS A1.

Config	Setup	Acc.	Δ A1	McN. p
A1	TFT only (Vertex reference)	51.79%	—	—
A4	+ Graph (proxy, deployed weights)	51.79%	± 0.00 pp	1.000
A8	A4 — gating (graph always used)	51.79%	± 0.00 pp	1.000
A2	+ Fundamental	DB-dependent		
A3	+ Sentiment	DB-dependent		
A5	+ Fundamental + Sentiment	DB-dependent		
A6	Seven-agent (deployed)	DB-dependent		
A7	A6 — data-quality rules	DB-dependent		
A9	A6 — weight redistribution	DB-dependent		
Graph-only (no TFT)		50.75%	−1.04 pp	0.271

TABLE VIII

WEIGHT-SENSITIVITY SWEEP FOR A4: VARYING THE ORCHESTRATOR’S GRAPH WEIGHT w_{GRAPH} WITH $w_{\text{TECH}} = 1 - w_{\text{GRAPH}}$, GATE ACTIVE. “FLIPS” COUNTS PER-ROW PREDICTIONS WHERE A4 DISAGREES WITH A1.

w_{graph}	Acc.	95% CI	McN. vs A1	Flips
0.00 (TFT only)	51.79%	[50.48, 53.12]	—	0
0.20 (deployed)	51.79%	[50.48, 53.12]	1.000	0
0.37 (re-norm.)	51.79%	[50.48, 53.12]	1.000	0
0.50	51.40%	[50.09, 52.74]	0.476	866
0.65	50.86%	[49.59, 52.17]	0.176	1,422

Falsifiable hypotheses, status on the offline subset. (i)

If $|A6 - A1| < 1$ pp the multi-agent layer is performance-neutral; for the offline-realizable subset {A1, A4, A8} the gap is 0.00 pp, consistent with the multi-agent layer contributing via robustness and auditability rather than accuracy. (ii) A8 = A4 exactly (no flips), so confidence gating contributes nothing to accuracy in this 2-agent ensemble; the gating-defense argument of Section V survives as an information-theoretic property (cannot be worse than A4) but is empirically inactive on this dataset. (iii) Per-segment breakdown (banks | insurance | non-banks) remains DB-dependent.

E. Economic Backtest of Baselines

Directional accuracy above the random floor does not guarantee exploitability: spread, volume, and execution friction can wipe the signal out, and on a thin market with a structural directional drift even a high accuracy can lose to buy-and-hold. The backtest-overfitting literature in quantitative finance has long documented how multiple-testing inflation and out-of-sample over-fitting produce strategies that look statistically significant in development and disappear in deployment [37], [38]. We reconstruct a pseudo-PnL series for every baseline by converting each row’s prediction into a 7-day position opened at close_t and closed at close_{t+7} , equal-weighted across active tickers, charging a 50 bps round-trip cost (conservative end of BVMT bid-ask + brokerage). We report the deflated

Sharpe ratio of [12] with $k = 9$ trials. Table IX reports the long-only NET regime (always-DOWN is flat under long-only and is omitted).

TABLE IX

LONG-ONLY ECONOMIC BACKTEST OF 8 BASELINES ON THE 2025 TEST SLICE, NET OF 50 BPS ROUND-TRIP COST, $r_f = 8\%$ (BCT TMM PROXY). DSR_p IS THE DEFLATED-SHARPE p -VALUE WITH $k = 9$; BOLD p -VALUES CLEAR 0.05. SORTED BY SHARPE.

Baseline	Cum.	Sharpe	Max DD	DSR _p
Standard Transformer	+38.7%	4.40	−4.6%	0.002
Simple LSTM	+38.0%	3.80	−12.1%	0.014
Logistic regression (L2)	+ 45.8%	3.69	−12.6%	0.021
Momentum 5/20	+47.8%	3.40	−12.7%	0.038
MA ₂₀ crossover	+38.1%	2.54	−15.5%	0.152
XGBoost	+20.4%	1.66	−17.0%	0.418
Always-UP	+28.0%	1.24	−27.3%	0.576
Random forest	+4.2%	−0.75	−10.1%	0.982

Three observations: (i) the 2025 test year had a strong positive drift (always-UP earns +28% cumulative net) so the binding benchmark is always-UP, not always-DOWN; (ii) four baselines clear the deflated-Sharpe bar (Standard Transformer, LSTM, LogReg L2, Momentum 5/20) but do so on a market with +28% realised drift, so the test year is favourable to long-biased strategies; (iii) the deep baselines concentrate their modest directional skill on days where the realised return is large, producing a markedly better Sharpe (4.40) and drawdown (−4.6%) than logistic regression despite indistinguishable pooled accuracy — a behaviour that raw accuracy alone misses. TFT v3’s economic profile on the harmonized slice is shipped at results/baselines/vertex_slice_harmonized_predictions and falls in the LSTM/Transformer range, consistent with the directional-accuracy ranking.

F. Five Observations

- **Trivial baselines are statistically indistinguishable.** Pairwise McNemar p -values among always-UP / always-DOWN / MA₂₀ crossover are all above 0.70. The four trivial / trend rows form a flat $\sim 50\%$ floor.
- **Tabular ML adds modest but real signal.** Logistic regression beats the trivial floor at 52.79% with the highest macro-F1 (0.524); McNemar against random forest gives $p = 1.6 \times 10^{-3}$.
- **Sequential deep baselines do not break the $\sim 53\%$ ceiling.** A vanilla two-layer LSTM and a standard 2-block Transformer reach 52.69% and 52.45% respectively on the same 60-day $\times$ 15-feature input; CIs overlap with logistic regression.
- **TFT v3 sits inside the same 50–53% band.** The reference TFT v3 (51.79%) is statistically indistinguishable from the strongest non-TFT baselines and from always-UP on the same slice. Two TFT ablations move the variant by ≤ 0.4 pp.
- **Thin-market implication.** No model in the panel produces a directional accuracy meaningfully separating from the slice’s 50% floor. The non-reproducible 77.4%

figure would have implied a skill exceeding the typical 55–65% TFT ceiling on developed-market data [1]; the reproducible 51.79% is consistent with the literature prior on emerging-market forecasting [24], [28], [38].

V. ROBUSTNESS, AUDITABILITY, AND EU AI ACT ALIGNMENT

The analyses in this section are *architectural*: we describe threat models and audit-trail properties that follow from the design choices, without running end-to-end poisoning experiments [10]. An empirical adversarial study including a SHAP-fooling baseline [25] is identified as future work.

Thin-market data poisoning surface. An adversary with modest capital can manipulate a single illiquid stock to affect global statistics. Per-stock target normalization isolates each stock i ’s (μ_i, σ_i) from manipulation of stock j ; per-encoder-window feature scaling additionally resets local statistics every $L = 30$ days. The attack surface is reduced to direct signal injection within a single window of a single stock, and confidence gating ($w_{\text{gated}} = w_{\text{orig}} \cdot \text{conf} / \tau$) provides a secondary defense by down-weighting low-confidence outputs in the ensemble. The ablation evidence (Section IV-B) shows these defenses impose no measurable accuracy cost on BVMT 2025.

XAI as audit trail. Every prediction produces a human-readable explanation describing per-agent contributions, feature attributions (SHAP for XGBoost, attention weights for TFT), and confidence breakdowns. These audit trails *provide the technical building blocks* for the transparency obligations of EU AI Act Articles 13–14 [6], while explicitly acknowledging that full conformity also depends on governance, risk management (Art. 9), data governance (Art. 10), technical documentation (Annex IV), post-market monitoring (Art. 72), and where applicable conformity assessment by a notified body (Art. 43). Empirical assumptions remain unvalidated: SHAP attributions are not formally tested for faithfulness on BVMT data, and adversarial robustness of explanations [25]–[27] is left to future work. We therefore phrase every regulatory claim as *aligned with* or *designed to support* the EU AI Act, not as “satisfying” or “conforming to” it.

VI. REPRODUCIBILITY AND ARTIFACT AVAILABILITY

The released artefacts support a one-command reproduction of every numerical claim in this paper. (i) **Training pipeline:** the Vertex-AI container (vertex/Dockerfile, vertex/cloudbuild.yaml) and the ablation runner (training/tft_ablations/runner.py) re-train the TFT v3 reference and the two internal ablations (TFT – VSN, TFT – per-stock norm) bit-for-bit from BVMT_SEED, in 71–161 minutes on a single NVIDIA T4. (ii) **Per-row predictions:** results/tft_ablations/vertex/baseline_test_predictions contains $(Q_{0.1}, Q_{0.5}, Q_{0.9}, r_7^{\text{true}})$ for every test row of the v3 reference and its two ablations. (iii) **Harmonized panel:** results/baselines/vertex_slice_harmonized_predictions contains per-row predictions for all ten non-TFT baselines plus the three TFT variants on the canonical $N = 5,582$ slice; the matching *.json carries

every number in Table V. (iv) **Multi-agent sweep:** `scripts/multiagent_offline_sweep.py` reproduces Tables VII and VIII in ~ 60 s on commodity CPU. (v) **Sentiment validation tooling:** `scripts/sample_news_for_labelling.py` and `scripts/evaluate_sentiment_labels.py` produce the 200-article expert-labelling sample and the Cohen’s κ evaluation report listed in the limitations.

The dataset is sourced from PostgreSQL tables populated from public BVMT price feeds (`histo_cotation*.csv`) and from the financial-statement and news scrapers in `scripts/scrape_ilboursa.py`; the post-cleaning TFT feature file (`data/features/tft_features.csv`, 105,191 rows, 13 MB) is released. Raw news articles and per-ticker financial-ratio rows depend on a database snapshot that is not yet published; this is the dominant remaining reproducibility limitation (Section VII).

VII. CONCLUSION

We present a reproducibility-first multi-component decision pipeline for thin-market equity forecasting on BVMT 2016–2025. The Vertex-reproducible TFT v3 reference reaches 51.79% directional accuracy (95% CI [50.43, 53.08]) on the harmonized 2025 test slice, statistically indistinguishable from ten classical and deep baselines and from always-UP. Two internal ablations move accuracy by ≤ 0.4 pp ($p > 0.65$), providing no significant evidence for or against the original architectural hypotheses. A multi-agent offline-realizable subset (A1, A4, A8) shows the graph signal cannot flip the binary technical decision under the deployed orchestrator weights; six remaining configurations are DB-dependent. An earlier internal estimate of 77.4% is documented as non-reproducible and retained as a transparency disclosure.

A. Limitations

- **Reproducibility of the historical 77.4% figure:** the dominant limitation. Without the original checkpoint we cannot fully discriminate between train/test leakage, a different test-set definition, and a different decision rule.
- **Multi-agent ablation coverage:** only 3 of 9 configurations realisable offline; 6 require the deployed PostgreSQL database (news pre-scores, financial ratios) and the trained fundamental checkpoint.
- **Sentiment validation:** the deployed sentiment model has not been validated on the BVMT corpus; per-article Cohen’s κ on a 200-article expert sample is shipped tooling but not yet executed.
- **Arabic coverage:** $\sim 15\%$ of news articles are Arabic and scored by a French-only model; their sentiment values should be treated as missing.
- **Quantile calibration:** 80% prediction intervals are not yet validated against empirical coverage rates.
- **Coverage of the 2025 test set:** the harmonized slice has 5,582 row-level predictions over 43 of the 68 tickers; performance on the absent 25 is unknown.

B. Future Work

- Replay A2, A3, A5, A6, A7, A9 once the database snapshot is published.
- Empirical poisoning study validating the architectural defenses of Section V.
- Quantile-calibration validation against empirical 80% coverage rates.
- AraBERT-based Arabic sentiment integration to lift coverage from 85% to 100%.
- Per-stock breakdown table with bootstrap CI and lift over always-UP on the harmonized slice (CSV shipped, table omitted for space).

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